

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Building Technology (AR246)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Candidates are required to give their answers in their own words.

The figures in the margin indicate full marks.

Marks: 100

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

- ✓ 1. What are the various components of the building and their functional requirement in the building?
2. Explain about various types of foundations with sketches. What are the various techniques of dewatering of foundation trenches? [4+4]
3. What are the requirement for a good partition wall? Draw a typical section of an outer wall showing all the details required for construction of wall. [4+4]
- ✓ 4. Design and draw plan of a 1.2 m wide doglegged stair for a residential building with 3m floor height.
- ✓ 5. What is roof used for? Sketch out different types of roof and show their parts. [3+5]
- ✓ 6. Explain the various measures taken for the thermal and sound insulation in a building.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

- ✓ 7. a. Define Shoring. What are the objectives of shoring? Describe the types of shoring with necessary sketches. [1+3+4]
b. Describe a septic tank and soak pit with necessary sketches. [7]
- ✓ 8. a. What are the various causes of foundation settlement? Define underpinning and explain different methods of underpinning with neat sketches. [3+5]
b. Why joints are necessary in the building construction? Describe in detail about the types of joints in structure and their uses. [2+5]

P.T.O.

9. a. What are the various non-destructive tests in the building structure?
What may be the causes of occurrence of cracks in the building and
what are the remedial measures to cracks? [3+5]
- b. Explain about the various components of the house wiring system.
[7]
10. Write Short Notes: (3X5=15)

- a. Damp Proofing in Wall
- b. Geo textile
- c. Solid and suspended Floor
- d. Fire Protection in Building
- e. Wall Finishing

11. Write an article on Smart and Energy efficient Building.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Hydraulics (CE247)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

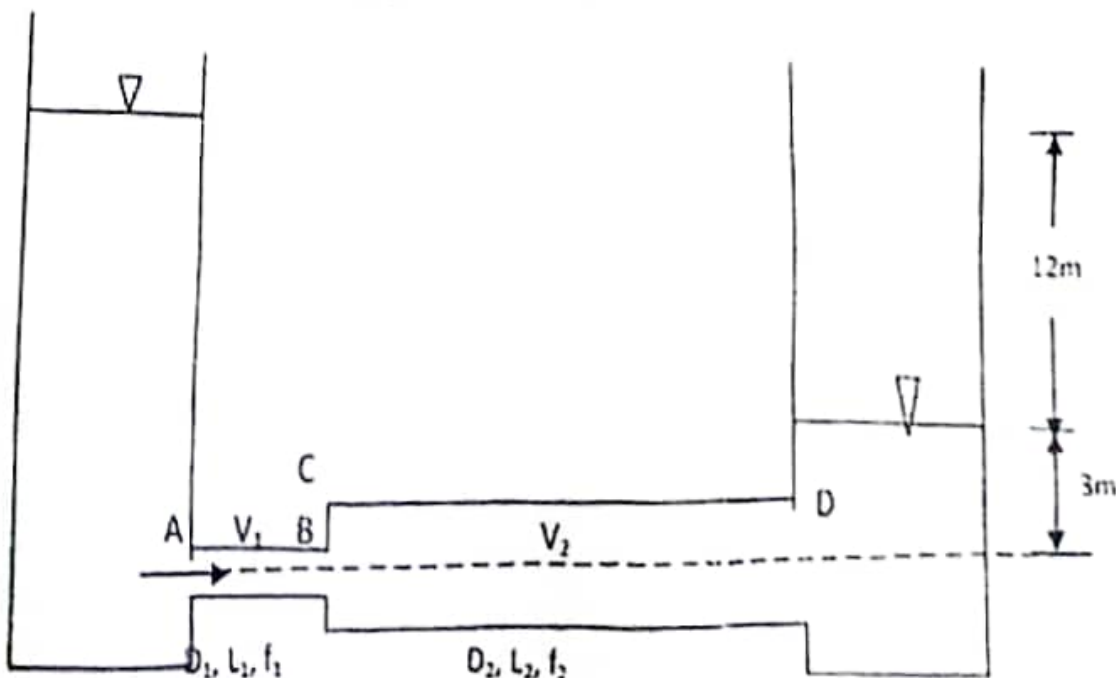
The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Oil of specific gravity 0.7 and viscosity 10 poise flows through a pipe of diameter 100mm. The velocity at the center is 2 m/s. Find
 - i. Mean velocity and radial distance at which this occurs
 - ii. Pressure gradient in the direction of flow
 - iii. Shear stress at the pipe wall
 - iv. Reynolds number
2. Compute the total loss of head in a reservoir combination whose surface elevations differ by 12m and are connected by a horizontal compound pipe system starting from the reservoir at higher level as shown in the figure.
 $L_1 = 150\text{m}$, $D_1 = 0.35\text{m}$, $f_1 = 0.007$
 $L_2 = 600\text{m}$, $D_2 = 0.8\text{m}$, $f_2 = 0.005$



3. a) What percentage of energy is lost in a hydraulic jump when $Fr_1 = 9$ and $y_1 = 0.4\text{m}$? [5]
 b) What is water hammer phenomena? Also point out its effects and remedial measures. [3]
4. a) What is the condition for the flow of maximum possible discharge in a rectangular channel for a constant specific energy? [5]
 b) Calculate the Mannings coefficient for a trapezoidal channel with side slopes 1:1 carrying a discharge of $1\text{m}^3/\text{s}$ at a depth of 1 m and laid at a slope of 0.001. [3]
5. A syphon of diameter 180mm connects two reservoirs having a difference in elevation of 18m. The length of the syphon is 750m and the summit is 4.5m above the water level in the upper reservoir. If the separation takes place at 2.5m of water absolute, find the maximum length of the syphon from upper reservoir to the summit. Take $f = 0.016$ and atmospheric pressure = 10.2m of water. Neglect minor losses.
6. a) Plot the possible types of GVF profiles for change in grades as 'Steep-Mild-Steep'. [3]
 b) Show the relation between sequent depths in a hydraulic jump using specific energy diagram and specific force diagram. [5]

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. Reservoir A, water surface elevation 120m is connected to reservoir B and C having surface elevation 70m and 50m respectively. A pipe line 150mm diameter and 400m long connects reservoir A to junction D. Reservoir B and C are connected to junction D by 75mm diameter 100m long and 100mm diameter 250m long pipeline respectively. Assuming friction factor $f = 0.04$ for all pipes, estimate the rate of flow for each pipe, neglecting minor head losses.
8. a) Derive the momentum equation for unsteady flow through pipes. [5]
 b) A lined rectangular channel with $n=0.015$ is 4.5m wide and has a flow depth of 2m with bed slope 1 in 1500. Keeping the channel section as rectangular and the perimeter constant, to what extent can the discharge be increased without changing the bed slope? [10]

- ✓ 9. A trapezoidal channel having a bottom width of 6m, side slope 2H:1V, $n=0.025$, and $s = 0.0016$ carries a discharge of $10\text{m}^3/\text{s}$. Compute the backwater profile created by a dam which backs up the water to a depth of 2.0m immediately behind the dam. Use direct step method for computation and solve in 3 steps.
- 10 ✓ a) Classify open channel flow. [5]
b) In a 3.5m wide rectangular channel laid at a slope of 1 in 3300, uniform flow occurs at a flow depth of 2m. Calculate how high a hump can be raised on the channel bed without causing a change in the u/s flow depth. If the u/s flow depth is to be raised to 2.4m, what should be the height of hump? Take $n = 0.014$. ✓ [10]
11. Write short notes on: [3+3+3+6]
a) Surge control
b) Types of hydraulic jump based on tail water depth
c) Venturi-flume for discharge measurement
d) Specific energy diagram

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Engineering Survey II (GE245)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Describe consecutive and independent coordinates. How you balance the consecutive co-ordinates. Explain the plotting method of traverse by grid coordinates. (2+2+4)
2. In order to carry out tachometric surveying, following observations were taken through a tachometer set up at station 'P' at height 1.235 m.

Staff held Vertical at	Horizontal distance from P	Staff Reading (m)	Angle of Elevation
Q	100	1.01	0°
R	200	2.03	0°
S	?	3.465, 2.275, 1.280	5°24'40"

Compute the horizontal distance of S from P and reduced level of station at S if R.L. of station P is 262.575m. (4+4)

3. What are the factors that affect the selection of contour interval? Explain the various characteristics of contour line with appropriate sketch. (3+5)
4. It is impossible to observe the length and bearing of line AB directly and the following are the observation made from two stations C and D.

Line	Length (m)	Bearing
CA	129.00	S 68°24' W
CD	294.00	N 20°36' E
DB	108.00	N 60°18' W

$\angle B = 180^\circ$
 $\angle B = 180^\circ$

5. How can you determine the depth of large water body? Explain the method of locating sounding positions. (3+5)
6. A down grade of 3.5% is followed by an upgrade of 4.5%. The reduced level and chainage of IP are 900.00m and 2450.00 m respectively. A vertical parabolic curve 180 m long is to be introduced to connect the two grades. The pegs are to-be-fixed-at 20 m intervals. Calculate the RLS of curve points including lowest point.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. a) State the principle of stadia tacheometry and describe the field method of finding tacheometric constants. (2+3)
- b) The top of temple was sighted from two stations-A and B at very different level. The observed vertical angle from A and B to top of temple 'P' were $30^{\circ}36'$ and $20^{\circ}12'$ respectively. The vertical angle B to A to a vane at 1.5 m above the foot of the vane was $4^{\circ}15'$. The height of instrument at A and B were 1.47 m and 1.42 m. The distance between two instrument stations was 112 m. RL of B was 1280.00 m. Find the RL of the top of the temple. Also apply the correction for refraction and curvature. (10)
8. a) What is degree of curve? Describe the elements of simple circular curve. (2+3)
- b) A road 8 m wide is to deflect through an angle of 60° with the Centre line radius of 300 m, the chainage of intersection point being 3605.00 m. A transition curve is to be used at each end of the circular curve of such a length that the rate of gain of radial acceleration is 0.5 m/s^3 , when the speed is 50 km/hr. Find out:
- Length of the transition curve.
 - Super elevation.
 - Chainage of all junction point.
- Lay out the transition curve by the method of deflection angles, taking peg interval as 5 m. (2+2+2+4)

9. a) What is analytical resection? Derive an expression to find the co-ordinates of unknown points by observations to three known points. (1+6)
- b) Co-ordinate of two points A and B are given below. A third point C has been chosen in such a way that bearing of AC and CB are $29^{\circ}30'$ and $45^{\circ}45'$ respectively. Calculate the length of the lines AC and CB. (8)

Point	Northing	Easting
A	150	200
B	1500	1300

10. a) A line 2350 m long laying at an elevation of 500 m measures 10.6 cm on vertical photograph. The focal length of camera used is 20 cm. Determine the scale of the photograph for an area having an elevation of 1200 m. (5)
- b) What is index contour? Explain the method of contour interpolation. Write the uses of contour map in civil engineering works. (10)
11. Write short notes on any three. (3x5=15)
- Function of GPS and its components and applications.
 - Remote sensing and its application.
 - Requirement of ideal transition curve and its equation determination.
 - Components of GIS and applications of GIS in Civil, engineering.
 - Expression of relief displacement.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Communication English II (SH241)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

- ✓ 1. Discuss how can listening be made effective.
- ✓ 2. Explain the major reading strategies to overcome effective reading problems.
- ✓ 3. Discuss the principles of effective speaking.
4. In a recent government policy, there was a significant increment of tax on imported electronic gadgets. You are the Sales Manager of Swift Communication, a local dealer in electronic devices. Due to the increase in import taxes, it is feared that the sales percent of such devices would drop. Write a memorandum to the sub-dealers of Swift Communication, informing them about the impact of new development (increased tax).
- ✓ 5. Draft a letter of recommendation to the General Manager of Universal Business House, Pokhara, in favour of Mr. T. R. Thapa for the post of Sales Manager who has worked in your organization for the last few years.
- ✓ 6. You are the Senior Consultant of Search Education, Mahendranagar. Draft a memo requesting all the employees of the branch to attend the welcome programme of a new employee at the certain date, time and venue.

P.T.O.

Group – 'B'

Attempt any four questions. (Long answer questions)

4×15=60

7. Assuming that you are the Representative of Geological Society, Kathmandu. Write a field report to the Director of the Institute on the recent discovery of copper mines in the area that you have recently visited. Invent necessary details about the topic.
- ✓ 8. Prepare a cover page, acknowledgement, conclusion and recommendations of a report on "The need of Preventive Measures for the Devastating Impact of Open Defecation in your Locality".
9. You want to initiate a Food Park somewhere in Mahendranagar that would function on the need of wonderfood. Write a brief proposal to the Regional Health Department, Dhangadi, Nepal.
- ✓ 10. Prepare the title page, table of contents, abstract, acknowledgements, conclusion and recommendation of the report entitled "Brain drain: Its reason and impact on the home country".
- ✓ 11. Choose a topic of your interest to carry out your research work. After you have identified the area, write an introduction and procedure for the dissertation.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Probability and Experimental Design (SH244)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

1. Define and distinguish between descriptive and inferential statistics with example. The scores of two golfers for 10 rounds each are

A:	58	59	60	54	65	66	52	75	69	52
B:	78	84	56	92	65	86	78	44	54	68

Which may be regarded as the more consistent player? Why?

(2+6=8)

2. Define and distinguish between conditional probability and Bayes theorem. There are three candidates for the position of Principal Mr. A, Mr. B and Mr. C, whose chance of selection are in 1:2:3 respectively. The probability that they will introduce co-education if they selected is 0.3, 0.5, and 0.8 respectively. Find the probability that there was co-education in the college. (2+6=8)
3. With examples describe the discrete and continuous random variable and their probability distributions. If the probability is 0.4 that steam will condense in a thin walled aluminum tube at 10 atm pressure, find the probability that under stated condition, steam will condense in 4 of 12 such tubes. (4+4=8)
4. Define and differentiate between normal distribution and standard normal distribution. An electrical firm manufactures light bulbs that have a life time, before burn-out, that is normally distributed with mean equals to 800 hours and standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours. (3+5=8)
5. With particular example discuss the meaning of two-dimensional random variable and its probability distribution. The joint probability function of two discrete random variables X and Y is given by $f(x, y) = \begin{cases} c(2x + y) & \text{where } x \text{ and } y \text{ are integer } 0 \leq x \leq 2, 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$. (i) find the constant c (ii) find $P(X \geq 1, Y \leq 2)$ (iii) find marginal function of X. (3+5=8)
6. With the help of an example related to your field discuss the meaning and application of estimation. A sample of 700 units showed that 200 were damaged. Find 99% and 95% confidence limits for the proportion of damaged units. (3+5=8)

P.T.O.

7. Define hypothesis and hypothesis testing. Is there any difference between estimation and hypothesis testing? Describe. The average weekly wages of a sample of 200 workers in industry A was Rs. 1150 with S.D of Rs. 100 and the weekly wages of a sample of 300 workers in industry B was Rs. 1050 with S.D of Rs. 50. Can we consider the weekly wages paid by industry A is higher than those paid by B at 5%? (2+2+1+10=15)

8. Define ANOVA test, describe its types and mention the assumptions of ANOVA test. The following are the breaking strength of three brand of cables

Brand	Breaking strength					
A	40	30	50	60	30	-
B	60	40	55	65	-	-
C	60	50	70	65	75	40

Construct ANOVA table and test for equality of average breaking strength of cables at 5%.

(2+2+3+8=15)

9. (a) Explain the concept of regression and correlation analysis with example and differentiate between them. The marks obtained by 10 students in statistics (X) and methodology (Y) are

X	52	50	55	50	60	70	72	78	80	63
Y	20	25	60	45	80	25	55	65	25	75

Calculate the rank correlation and interpret it.

(2+6=8)

- (b) A population consists of four numbers 3, 7, 11, 15. If samples of size two are drawn without replacement, show that sample mean is unbiased estimator of population mean. (7)

10. (a) State the assumptions for using Z-test and t-test in hypothesis testing. It is claimed that both tea and coffee are equally popular. A random sample of 1200 person 650 are regular consumers of tea. Is the claim justified at 5% level of significance? (3+5=8)

- (b) Define probability density function and probability distribution function. If a random variable X has function $f(x) = \begin{cases} 2e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$. (i) verify that function is probability density. (ii) find $P(1 < X < 3)$ (iii) $P(X > 0.5)$. (2+5=7)

11. (a) With example discuss the meaning of Hypergeometric and negative binomial distribution. A boy is throwing stone at a target. What is the probability that his 10th throw is his 6th hit, if the probability of hitting the target is 0.6? (4+4=8)

- (b) The shear resistance of soil y KN m⁻² is determined by measurements as a function of the normal stress, x KN m⁻² is as follow

Stress(X)	10	11	12	13	14	15	16	17	18
Resistance(Y)	14.08	15.57	16.94	17.68	18.49	19.55	20.68	21.72	22.80

Estimate the shear resistance when normal stress is 20 KN m⁻².

(7)

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Structural Analysis I (CE243)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt any five questions. (Short answer questions)

5×8=40

- ✓ 1. Define linearly elastic structures and differentiate between dynamic and static load analysis in the structure. [3+5]
2. Deduce the relation for a unit load method using the virtual work principle for a beam element.
- ✓ 3. Determine the vertical displacement of Joint C as shown in fig. 1 by strain energy method. EI is Constant throughout.

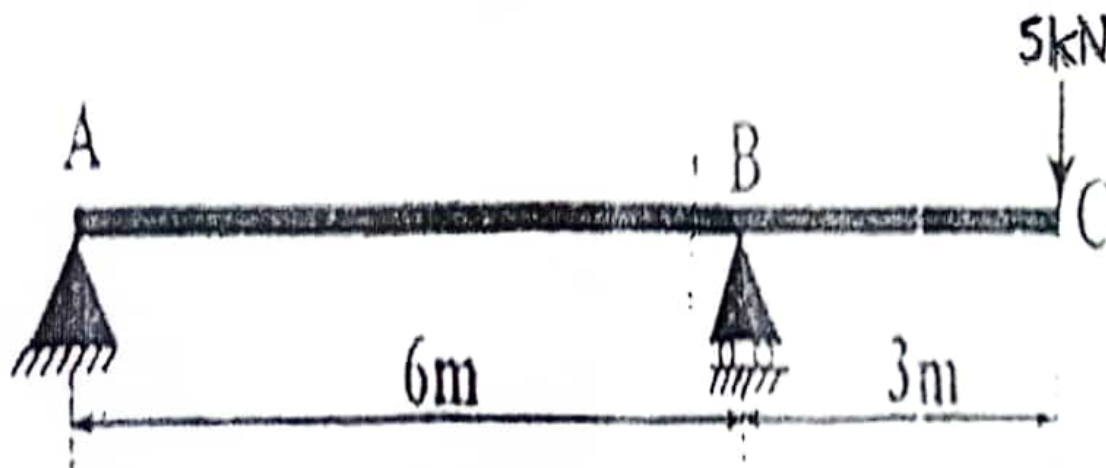


Fig. 1

4. A uniformly distributed load of intensity 30 kN/m crosses a simply supported beam of span 60 m from left to the right. The length of the UDL is 15 m . Find the maximum bending moment for a section 20 m from the left end. Find also the absolute maximum bending moment and shear force in the beam. [4+4]
5. A three-hinged symmetrical parabolic arch of span 50 m and rise 5 m is subjected to the rolling loads. Determine the maximum moment at a section 20 m from the right support due to the concentrated load of 20 kN at 20 m from the left support and a uniformly distributed load of 25 kN/m distributed over the right half of the arch.
6. A simply supported beam of span 6 m with an overhang of length 2 m on both sides is loaded at the free ends with a concentrated load of 20 kN each, besides UDL of 10 kN/m in the middle third of the span. Calculate the deflection of mid span using Macaulay's method.

Group – 'B'

Attempt any four questions. (Long answer questions)

$4 \times 15 = 60$

7. Determine the deflection at free end of the beam shown in the fig 2. using moment area method or conjugate beam method

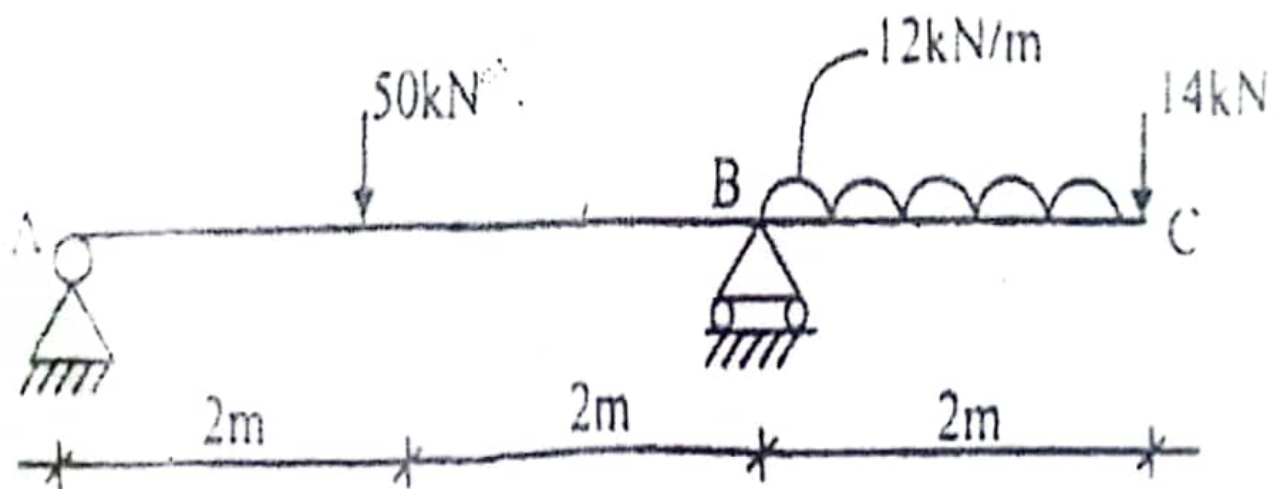


Fig.2

0.75
1.25

8. In the truss shown in figure below, determine the vertical displacement of joint H and horizontal displacement at joint B. The rise in temperature in all top chord members is 60°C . The cross sectional area of all members is 300 mm^2 . Take $E = 2 \times 10^5 \text{ N/mm}^2$ and coefficient of thermal expansion is $12 \times 10^{-6}/^{\circ}\text{C}$

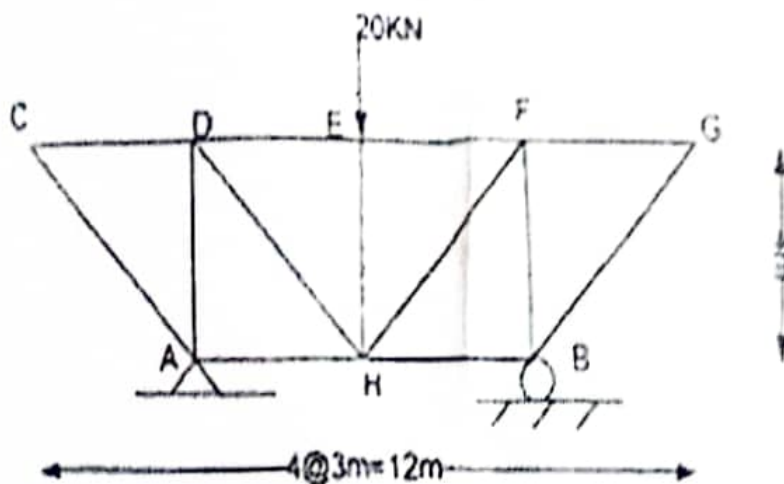


Fig. 3

9. What do you mean by influence line diagram? The following system of concentrated loads rolls from left to right on the girder of span 60 m, with 40 kN load leading. Determine the absolute maximum moment, absolute maximum shear force and maximum moment and maximum shear force at 15 m from the left support.

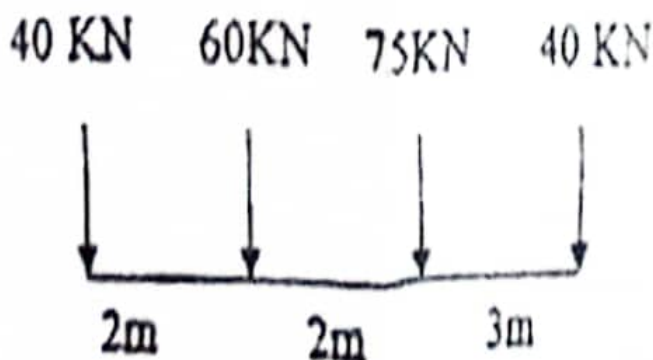
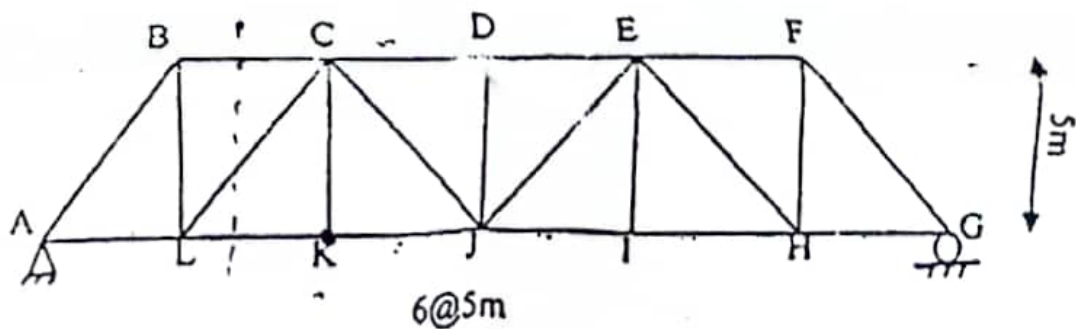


Fig. 4

10. (a) Draw an ILD for equivalent cable load we in three hinged stiffening girder of span L . [7]
- (b) The cables of a suspension bridge having a span of 100 m and central dip of 10m. Each cable is stiffened by a girder hinged at two ends and at mid span. The dead load is a uniformly distributed load of 10 KN/m for the entire span. The live load is uniformly distributed load of length 20 m and intensity 30 KN/m. Determine the maximum tension in the cable when the live load is placed on the left half of the span with the leading edge at the central hinge. Determine the bending moment and shear force in the girder at 25 m from the left support. [8]
11. Find the ILD for the member AB, BC and LK as shown in fig. 5 when the load moves in the bottom chord.



30 Fig. 5